

Do Small Vision-Language Models Route Language the Same Way in Pixels and in Tokens? A Mechanistic Investigation with Code-Switched Text

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Abstract

Vision-language models (VLMs) often read text from images, including code-switched text such as a screenshot of a Hinglish tweet. It is unknown whether the attention heads that matter for a language when it arrives as text tokens also matter when it arrives as a rendered image. We study this question in four open VLMs with at most 3B parameters. We score every attention head with gradient-based head importance (LAHIS) on monolingual English, Spanish, and Hindi sentences. Each sentence is given as text, as a rendered image with a query, and as a rendered image alone. We also probe the vision encoders for script identity and ablate the top heads while the models read Hindi-English code-switched images. The answer depends on the model. Qwen2-VL-2B and SmolVLM use largely different heads for text and image input. InternVL3-2B uses nearly the same heads, but only for Hindi. For PaliGemma-3B, the overlap measure cannot separate either pattern from chance. All four vision encoders encode script identity from their first layer. Zero-ablating the top 20 heads for either language destroys generation in every model, while ablating 20 random heads does not. The top heads are therefore necessary for coherent output, but the ablation does not show that they route language. A script-change metric counts this collapse as a change of script, so head-ablation studies need a check on output quality. We release the pipeline, the corpus, and the per-head scores.

1 Introduction

VLMs read text from images such as screenshots, photographed notes, and scanned documents. This text is often code-switched, which means it alternates between two languages within one sentence. A VLM must then identify both languages from pixels, without the token boundaries that a text-only model receives.

A VLM passes an image through a vision encoder and sends the resulting tokens to a language decoder. The same decoder also reads text tokens directly. Mechanistic studies have located the attention heads and neurons that matter for language processing in text-only models (Liu et al., 2026; Tang et al., 2024). Studies of VLMs have located heads that read text in images, but only for English (Baek et al., 2025). It is unknown whether the decoder relies on the same heads for a language when it reads that language as text tokens and as image tokens.

The answer determines whether findings about language heads in text models transfer to visual input. If both input paths use the same heads, text-only findings apply to images. If they use different heads, the visual path needs its own analysis.

We compare the two paths with three input conditions. Each sentence is given as text (condition A), as a rendered image with a fixed query (condition B), and as a rendered image alone (condition C). Comparing A with C isolates the input modality. Comparing B with C isolates the effect of the query. We score every attention head in each condition with LAHIS (Liu et al., 2026) and study four open VLMs from three decoder families. We ask two research questions and one exploratory question:

- **RQ1.** When the same sentence reaches a VLM as text tokens and as a rendered image, does the model rely on the same attention heads?
- **RQ2.** When a VLM reads code-switched text from an image, does ablating the top heads change the language of its output more than ablating random heads?
- **Exploratory.** Do the positions of the top heads match across models that share a decoder family?

We first test a precondition. The vision encoder must represent script identity (Latin or Devanagari) for the image path to route language by script.

Following Liu et al. (2026), we identify heads on monolingual sentences. Code-switched text enters the study in two places. The script probes read mixed-script images, and the ablation test in RQ2 removes the identified heads while the model reads code-switched images.

We find the following:

- **Script identity is available early.** All four vision encoders separate Latin from Devanagari script from their first layer (Section 6.1).
- **Head sharing depends on the model.** Two models use largely separate heads for text and image input, one shares them for Hindi only, and one is inconclusive. Where the sets differ, the text heads concentrate in the first two decoder layers and the image heads do not (Section 6.2).
- **The top heads are necessary but not shown to route language.** Zero-ablating them destroys generation in all four models, while random heads do not. The ablation cannot isolate a role in routing language (Section 6.3).
- **Shared positions across models match layer placement.** The two Qwen-family models share about as many head positions as their layer placement predicts (Section 6.4).

We study four small VLMs, two language pairs, and one attribution method. We report results per model and rate the evidence for each observation (Section 7). We make no claim that the patterns hold for other VLMs.

2 Related Work

Language heads and neurons in text LLMs. Liu et al. (2026) introduce LAHIS, which scores attention heads with one forward and one backward pass through a differentiable head mask. They find that a small set of heads matters most for language identification in multilingual LLMs. Tang et al. (2024) identify language-specific neurons by contrasting activations on parallel corpora. Deactivating these neurons degrades only the corresponding language. Wendler et al. (2024) show that intermediate representations of Llama-2 are closer to English than to the input or output language. All of this work studies text input only.

Language confusion and code-switching. Nie et al. (2025) trace language-confusion errors in English-centric LLMs to specific token positions and late-layer transitions. We know of no mechanistic study of code-switching itself, and of none that covers visual input.

Language heads in VLMs. Ye et al. (2025) identify cross-lingual attention misalignment in VLMs and intervene on it to reduce multilingual object hallucination. They study monolingual queries in several languages and do not separate the text path from the image path. Baek et al. (2025) identify OCR heads that read text in images, but only for English text.

Reading text from pixels. Several models process rendered text directly as pixels (Rust et al., 2023; Tschannen et al., 2023) or read documents and screenshots without an OCR step (Kim et al., 2022; Lee et al., 2023). CLIP’s image encoder matches images of words with images of the scenes they describe, and its spelling ability can be isolated in a separate subspace (Materzyńska et al., 2022). Many attention heads in CLIP’s image encoder have property-specific roles, such as location or shape (Gandelsman et al., 2024). Our script probes test a narrower property, script identity, in the vision encoders of VLMs.

Visual information in the decoder. Causal tracing has been adapted to VLMs to locate where image information is stored and transferred (Palit et al., 2023; Basu et al., 2024). Neo et al. (2025) track how visual token representations change across the layers of LLaVA’s language model. None of these studies compares the heads used for the same content as text and as an image.

Gap. No prior work compares, head by head, which heads a VLM relies on for the same sentence as text and as a rendered image. Our design also separates the effect of the query from the effect of modality.

3 Data

All code-switched text is real and human-produced, with gold token-level language labels. We do not use synthetic code-switching, because word substitution can produce sentences that bilingual speakers would not write.

The Hindi-English data are the Devanagari-script Twitter portion of the GLUECoS part-of-speech corpus (Khanuja et al., 2020). The Spanish-English data are the code-switched sentences of the Bangor Miami corpus of transcribed speech (Deuchar et al., 2014). Monolingual English, Spanish, and Hindi sentences come from FLORES-200 (NLLB Team et al., 2022).

Cleaning. We apply Penn Treebank detokenization and filters for script consistency and noise. A two-stage fluency filter first applies heuristic checks and then removes the quartile with the highest Qwen2.5-0.5B perplexity in each language pair. A content filter removes profanity and political material, judged by Claude against a fixed rubric. We record the reason for every removed sentence. The final corpus has 806 Hindi-English sentences from the GLUECoS training and development splits, 127 Spanish-English sentences, 750 monolingual sentences (250 per language), and 100 transliteration controls. The controls are Hindi sentences written in Latin script, so they separate script from language.

Data for each analysis.

- **Head importance (RQ1).** For each language, 100 of the 250 monolingual FLORES-200 sentences. Conditions B and C use rendered images of the same sentences. No code-switched text is used to identify heads.
- **Script probes.** The first 300 rendered code-switched images, 127 Spanish-English and 173 Hindi-English. All Devanagari patches therefore come from the Hindi-English images.
- **Hindi-English ablation (RQ2).** 100 sentences sampled from the 806 Hindi-English sentences. These are disjoint from the sentences used to identify heads.
- **Spanish-English ablation.** The first 100 of the 127 Spanish-English sentences.

Rendering. Each sentence is kept as raw text and rendered as black text on a white image. Each token is drawn in a font for its script, Noto Sans for Latin and Noto Sans Devanagari for Devanagari. We record a bounding box for every token, which gives each image patch a script label. Figure 1 shows examples.

4 Models

We study four open VLMs that differ in vision encoder, decoder family, and image-token compression. Each has at most 3B parameters, which keeps gradient attribution over every head within our compute budget.

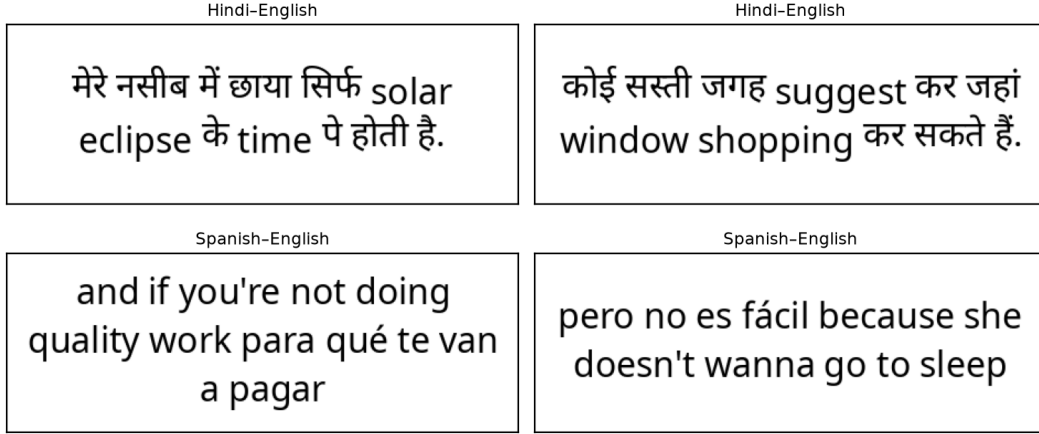


Figure 1: Hindi-English stimuli mix Devanagari and Latin script within one image, while Spanish-English stimuli use Latin script only. The four sentences are rendered as the models receive them, without a prompt.

- **Qwen2-VL-2B-Instruct** (Wang et al., 2024) pairs a custom 2D-RoPE vision transformer (0.67B) and a 2×2 patch merger with a Qwen2-1.5B decoder (28 layers, 12 heads per layer).
- **InternVL3-2B** (Zhu et al., 2025) pairs InternViT-300M-448px-V2.5 and pixel-shuffle compression with a Qwen2.5-1.5B decoder (28 layers, 12 heads per layer).
- **SmolVLM-Instruct** (Marafioti et al., 2025) pairs a SigLIP-SO400M encoder (Zhai et al., 2023) and a pixel-shuffle connector with a SmolLM2-1.7B decoder (24 layers, 32 heads per layer).
- **PaliGemma-3B (mix-448)** (Beyer et al., 2024) pairs a SigLIP-SO400M encoder at 448 pixels and a linear projector with a Gemma-2B decoder (18 layers, 8 heads per layer, multi-query attention). The projector keeps all 1,024 image tokens. PaliGemma-3B is a prefix language model, so we score its decoder on the text after the prefix `[BOS] \n`.

Qwen2-VL-2B and InternVL3-2B use decoders from the same family. Qwen2.5 was pretrained on 18 trillion tokens, against 7 trillion for Qwen2 (Qwen Team et al., 2024). The Qwen2.5 report does not state whether Qwen2.5 was initialized from Qwen2 weights. SmolVLM uses a Llama-architecture SmolLM2 decoder, and PaliGemma-3B uses a Gemma decoder. Both come from separate pretraining lineages.

Language competence. We check that each extracted decoder models each language before we score heads, so that the scores do not reflect noise. Table 1 reports decoder perplexity on monolingual FLORES-200 text. Per-token perplexity cannot be compared across languages. Tokenizer fertility differs by 3.4 to 3.6 times between English and Devanagari Hindi, which lowers Hindi’s per-token perplexity regardless of competence. We therefore normalize by characters, $\text{ppl}_c = \exp(\text{NLL}/n_{\text{chars}})$, and use this value for all cross-language comparisons. All models stay below a per-token threshold of 100 except PaliGemma-3B on Spanish (114.7). We keep PaliGemma-3B because its character-normalized Spanish perplexity lies within the range of the other models.

5 Method

Figure 2 summarizes the pipeline.

5.1 Input conditions

Each sentence is scored in three conditions:

Table 1: Every decoder models all three languages, and Hindi is the hardest language for every model on the character-normalized scale. Per-token perplexity is lower for Hindi because of tokenizer fertility and is not comparable across languages. PaliGemma-3B is scored as a prefix language model.

Model	Per-token ppl.			Char.-normalized ppl.		
	en	es	hi	en	es	hi
Qwen2-VL-2B	40.9	26.2	5.9	2.06	2.23	4.46
InternVL3-2B	38.4	28.0	6.7	2.04	2.28	4.87
SmolVLM	83.8	49.5	7.4	2.23	2.98	7.64
PaliGemma-3B	91.3	114.7	69.4	2.48	2.75	4.48

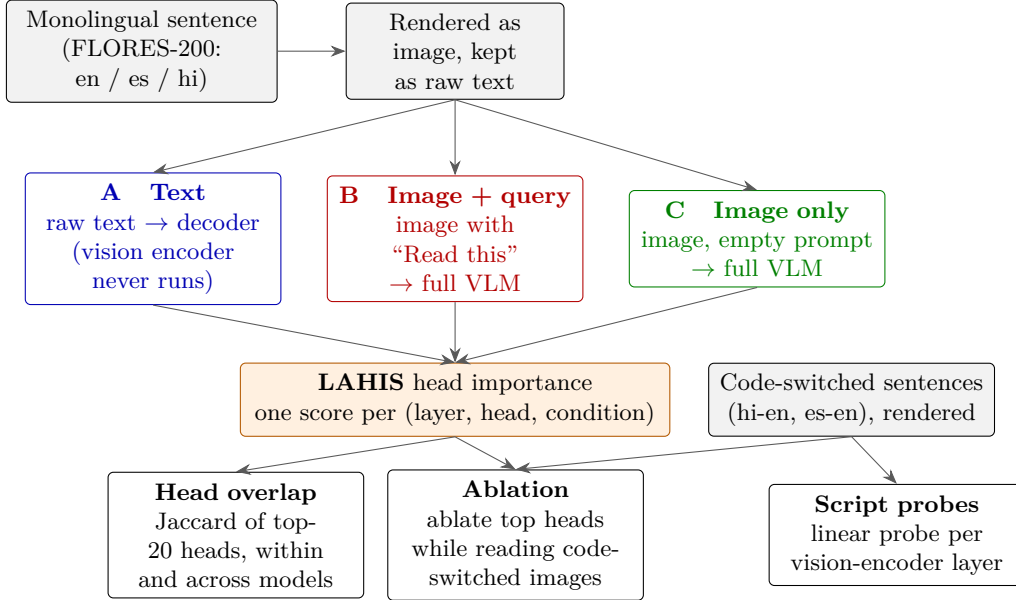


Figure 2: Head importance is scored on monolingual sentences in three input conditions and compared across conditions and models. Rendered code-switched sentences are used for the script probes and for the ablation test.

A (text) The decoder reads the raw sentence. The vision encoder does not run.

B (image+query) The full VLM reads the rendered sentence with the fixed query “Read this:”. The loss is computed on the sentence as the target.

C (image-only) As in B, but the query is empty. The chat template and role tokens are identical to B.

Comparing B with C isolates the effect of the query. Comparing A with C isolates the effect of modality. Condition A uses the decoder taken from inside the VLM after vision-language training, not the original pretrained checkpoint. This keeps the comparison between A and C free of differences caused by vision-language training.

5.2 Head importance

Following Liu et al. (2026), we multiply the output of each head h in layer ℓ by a mask value $M_{\ell,h}$, initialized to one. We apply the mask at the input of each layer’s attention output projection, where the head outputs are still separate. The importance of a head is $|\partial\mathcal{L}/\partial M_{\ell,h}|$, where $\mathcal{L}$ is the language-modeling loss over all tokens of a sentence. We average this score over 100 sentences per language and condition. Appendix E shows the resulting layer-by-head maps.

What the score measures. The LAHIS loss is the standard next-token loss and does not target language identity. A head scores high if masking it hurts prediction of text in the given language. This includes heads with general functions, such as tracking position or copying earlier tokens. A high score therefore marks a head as important for processing a language, including language-general heads. Section 6.4 shows that most top heads are shared across the three languages.

5.3 Head overlap

For each model, language, and condition, we take the 20 highest-scoring heads and compute the Jaccard overlap between conditions. The chance level is the 95th percentile of the overlap between random 20-head sets, over 10,000 draws from each model’s grid. We repeat the analysis for $K \in \{5, 10, 20, 40, 80\}$ heads. For the cross-model analysis, we count the (layer, head) positions that two models’ text-condition sets share.

5.4 Script probes

At every vision-encoder layer, we train a linear probe to predict whether an image patch shows Latin or Devanagari script. Patch labels come from the token bounding boxes, mapped onto the grid of image tokens that the decoder receives. This grid differs from the raw patch grid when a patch merger or pixel shuffle compresses the tokens. We exclude background patches, so chance accuracy is 0.5. We sample 100 patches per image and 4,000 text patches per layer. The probe is a class-balanced logistic regression, evaluated by balanced accuracy with stratified 5-fold cross-validation. The folds split patches, not images (Section 8).

5.5 Ablation

Gradient scores are correlational, so we test whether the top heads affect generation. We set the output of a head set to zero and generate from 100 Hindi-English code-switched images, with greedy decoding and a limit of 64 tokens. The ablated sets are the 20 highest-scoring text-condition heads for English and for Hindi. We call them English heads and Hindi heads after the language used to score them. The sets are not contrasted between languages, and they overlap heavily (Section 6.4).

We use two controls. The random-head control ablates 20 heads drawn from outside both top-20 sets. Mean ablation replaces each head’s output with its mean over token positions in the same image’s un-ablated forward pass. We run the ablation once and then three more times with new seeds. Each seeded run draws its own 100 sentences and its own random heads.

We measure two outcomes. The first is the script label of each output, which is Latin, Devanagari, other, or none for output without letters. A script change is any change of label from the un-ablated output. The second is fluency. An ablated output counts as fluent if it keeps at least half of the words of the un-ablated output for the same image. This proxy is coarse, so we compare it against the random-head control. Spanish-English uses one script, so for it we run a language identifier (`langid`) on the output instead.

6 Results

6.1 Script identity is decodable from the first layer of every vision encoder

All four vision encoders separate Latin from Devanagari patches above chance from layer 0 (Figure 3, Table 2). Accuracy varies widely across encoders. InternVL3-2B and PaliGemma-3B reach near-perfect accuracy, Qwen2-VL-2B plateaus near 0.81, and SmolVLM stays near 0.63 at every depth.

The same encoder family can give very different results. PaliGemma-3B and SmolVLM both use SigLIP-SO400M, but they sit at opposite ends. PaliGemma-3B runs the encoder at 448 pixels and keeps all 1,024 patch tokens, while SmolVLM gives it lower-resolution, pooled tokens. All models see the same images and fonts, so this deployment difference is a likely cause.

The probes track script rather than language. Each model’s best-layer probe labels most transliteration controls as Latin, at rates at or above its accuracy on code-switched images (Table 2). These controls are

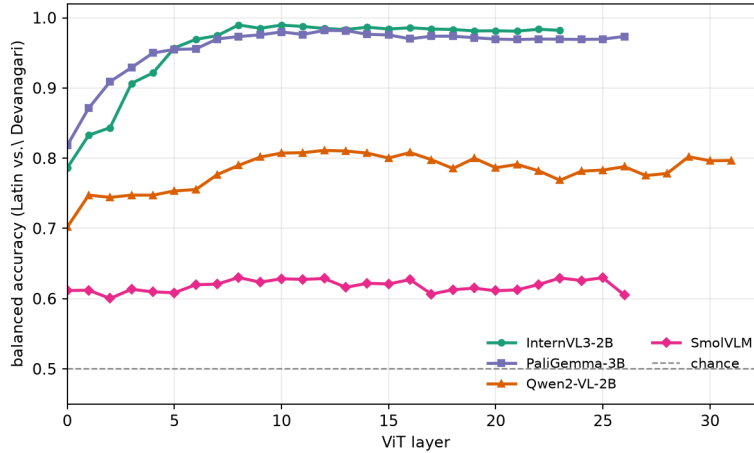


Figure 3: Script identity is linearly decodable from the first vision-encoder layer in all four models, but accuracy ranges from near chance to near perfect. SmolVLM does not improve with depth. The dashed line marks chance (0.5).

Table 2: Every encoder separates the two scripts above chance, and every probe labels Hindi written in Latin script as Latin. Peak and code-switched columns give balanced accuracy, which patch-level cross-validation may inflate (Section 8). The last column gives the share of transliteration controls labeled Latin. It uses new images and is not affected.

Model	Encoder	Peak (layer)	Code-switched (%)	Translit. as Latin (%)
InternVL3-2B	InternViT-300M	0.990 (8 of 24)	98.8	99.7
PaliGemma-3B	SigLIP-SO400M	0.982 (12 of 27)	98.2	99.7
Qwen2-VL-2B	custom ViT	0.811 (12 of 32)	80.2	97.3
SmolVLM	SigLIP-SO400M	0.630 (8 of 27)	62.4	84.0

new images, so the patch-level cross-validation does not affect this check. Script identity is therefore available to every decoder. Differences between text and image head sets (Section 6.2) do not come from a missing script signal.

6.2 Whether text and image inputs share heads depends on the model (RQ1)

Figure 4 compares two overlaps for each model and language. The first is between the text and image-only head sets ($A \cap C$). The second is between the two image conditions ($B \cap C$). Table 5 in Appendix A lists all values. The four models show three patterns.

Separate heads in Qwen2-VL-2B and SmolVLM. In both models, text-image overlap stays low for every language, while the two image conditions overlap strongly. Once an image is present, these models use a consistent head set, with or without a query. That set differs from the head set for the same sentence as text.

Shared heads for Hindi in InternVL3-2B. InternVL3-2B uses nearly the same heads for Hindi as text and as an image. Its Hindi text-image overlap is 0.667, well above its English and Spanish overlaps. Hindi competence does not explain this result. Qwen2-VL-2B models Hindi at least as well (Table 1) and keeps its heads separate. With one model and one language, we treat this result as tentative.

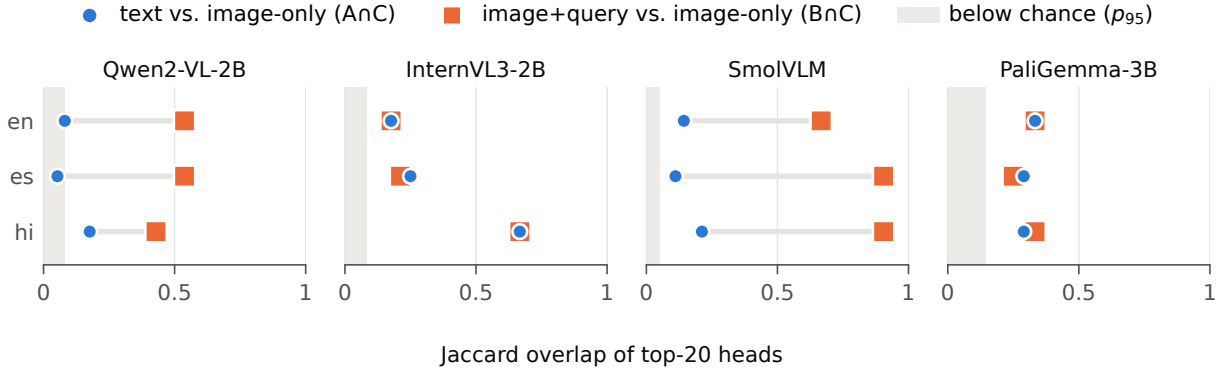


Figure 4: Text and image inputs use separate heads in Qwen2-VL-2B and SmolVLM, nearly the same heads in InternVL3-2B for Hindi, and an unresolved pattern in PaliGemma-3B. Circles show text versus image-only overlap ($A \cap C$). Squares show image+query versus image-only overlap ($B \cap C$). Shading marks each model’s chance level.

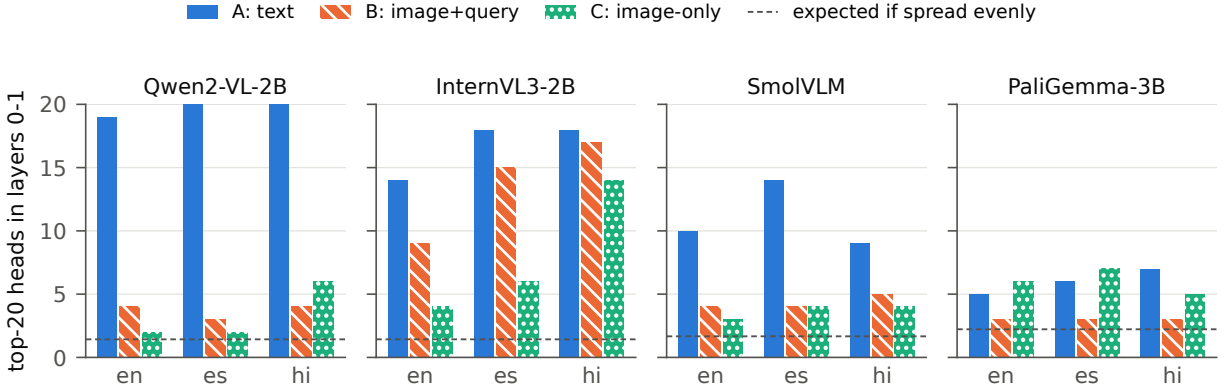


Figure 5: Text-condition heads concentrate in layers 0–1 in the two Qwen-family models, while image-condition heads mostly lie in later layers. InternVL3-2B’s Hindi image-condition heads are the exception. The dashed line shows the count expected if the 20 heads were spread evenly over the grid.

Inconclusive overlap in PaliGemma-3B. All of PaliGemma-3B’s overlaps lie between 0.21 and 0.33, modestly above its chance level of 0.143. Its image-image overlap is not higher than its text-image overlap. Its grid has only 144 heads, so a top-20 set covers a large fraction of it and Jaccard overlap loses resolution.

Layer placement. The layer placement of the top heads matches these patterns (Figure 5, Table 7). Qwen2-VL-2B places almost all of its text-condition heads in layers 0–1 but few of its image-condition heads there. InternVL3-2B also places its text-condition heads early. For Hindi, its image-condition heads are early as well, which matches its shared Hindi heads. SmolVLM places its text-condition heads partly early and its image-condition heads mostly later. PaliGemma-3B places few heads in layers 0–1 in any condition.

Robustness. The three patterns hold for every K from 5 to 80 (Appendix B).

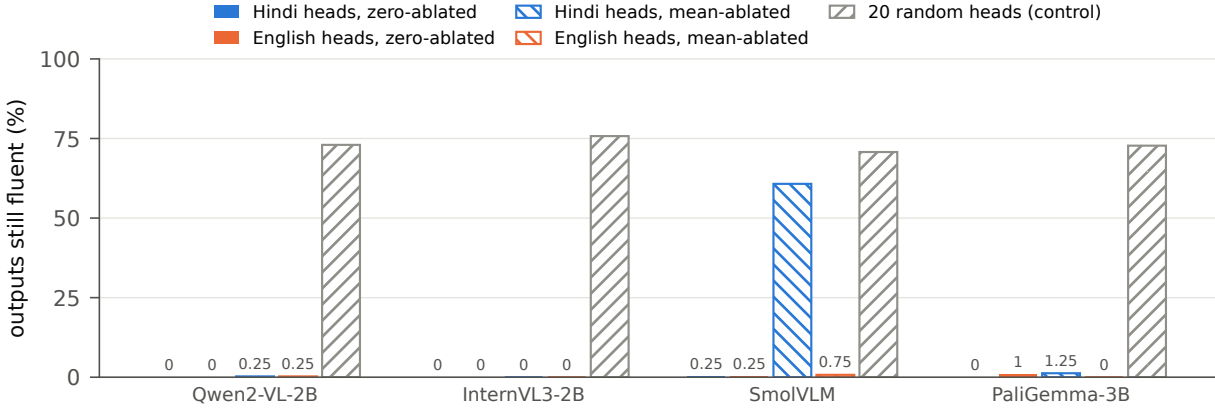


Figure 6: Zero-ablating the top 20 heads for either language destroys generation in every model, while ablating 20 random heads leaves most outputs fluent. Mean ablation also destroys generation, except for SmolVLM’s Hindi heads. Bars show the percentage of 400 generations that keep at least half of the un-ablated output’s words.

Encoder signal and shared heads. A strong script signal in the encoder does not by itself lead to shared heads. InternVL3-2B and PaliGemma-3B have the strongest script signals, but only InternVL3-2B shares heads. PaliGemma-3B’s overlap is inconclusive, so this comparison is weak.

6.3 The top heads are necessary for coherent output, but ablation does not isolate language routing (RQ2)

Zero-ablating the top 20 heads for either language destroys generation in every model (Figure 6). At most 1% of outputs stay fluent, against 71–76% for the random-head control. The collapse takes a different form in each model. PaliGemma-3B mostly returns empty strings. Qwen2-VL-2B returns output without letters or with fragments of Chinese, Japanese, or Korean (CJK) characters. InternVL3-2B’s Hindi-head ablations contain CJK characters in every output, and SmolVLM repeats a special token. No un-ablated output contains CJK characters. Mean ablation also destroys generation, except for SmolVLM’s Hindi heads. Appendix D gives the counts.

The top heads are therefore necessary for coherent output, and a random set of the same size is not. The ablation does not show that these heads route language. English heads and Hindi heads cause the same collapse, and the two sets largely coincide (Section 6.4). In Qwen2-VL-2B they cover almost all 24 heads in layers 0–1, so the collapse may reflect general damage to the first layers. A test of language routing needs a random control drawn from the same layers and head sets contrasted between languages.

A script-change rate misreads this collapse as a change of language. Output without letters changes the script label and therefore counts as a script change. In the single run, all 74 script changes after ablating Qwen2-VL-2B’s Hindi heads were outputs that lost all letters, and so were all 97 for PaliGemma-3B. InternVL3-2B shows a low rate because its collapsed outputs keep enough Latin letters to retain a Latin label. Differences in script-change rate between models therefore reflect how each model fails, not how it routes language. Appendix D reports the script-change rates. The Spanish-English rates, based on a language identifier, have not been checked for collapse.

Without ablation, SmolVLM produces Devanagari output for 37 of 100 images, against 5 to 15 for the other models. Competence does not explain this, because SmolVLM has the highest Hindi perplexity (Table 1). We leave its cause open.

Table 3: The two Qwen-family models share the most head positions, close to the number expected from layer placement alone. The expectation assumes that each model places its heads in layers 0–1 at random within those two layers.

Pair	en	es	hi
Qwen2-VL-2B, InternVL3-2B	13	16	15
expected from layer placement	11	15	15
Qwen2-VL-2B, PaliGemma-3B	5	6	7
InternVL3-2B, PaliGemma-3B	5	6	7
Qwen2-VL-2B, SmolVLM	2	2	1
InternVL3-2B, SmolVLM	1	1	1
SmolVLM, PaliGemma-3B	0	0	0

6.4 Shared head positions across models match layer placement (exploratory)

Most top text-condition heads are shared across languages within each model. The number of heads in the top 20 for all three languages is 18 for Qwen2-VL-2B, 14 for InternVL3-2B, 14 for SmolVLM, and 15 for PaliGemma-3B. LAHIS therefore mostly ranks language-general heads. The English and Hindi sets ablated in Section 6.3 are largely the same heads.

Across models, the two Qwen-family models share the most head positions (Table 3). The other pairs share far fewer, and SmolVLM and PaliGemma-3B share none.

Layer placement predicts the overlap of the Qwen-family pair. Both models place most of their top text-condition heads in layers 0–1, which hold 24 heads in each model. If each model placed those heads at random within the two layers, the sets would share about 11, 15, and 15 positions for English, Spanish, and Hindi. The observed counts are 13, 16, and 15. The other models place fewer top heads in layers 0–1 and have differently shaped grids, so they share fewer positions.

A shared head position has no meaning beyond layer placement unless the two models share initialization. Heads within a layer are interchangeable across independently trained models. The Qwen2.5 report does not state whether Qwen2.5 was initialized from Qwen2 weights (Section 4), and the other pairs come from separate lineages. The data therefore show a layer-level pattern. With four models, decoder family varies together with head count, grid shape, attention type, and image compression. Our data cannot show whether the pattern follows decoder family.

7 What We Can and Cannot Conclude

Table 4 rates the evidence for each observation. *Supported* means consistent evidence against a matched control. *Tentative* means evidence from one model or one language. *Inconclusive* means the measure cannot separate the alternatives. *Not supported* means the evidence does not back the claim.

Two points follow for interpretability research on VLMs. First, language heads found in one VLM may not transfer to another. Our four models differ in whether text and image inputs share heads, in where their top heads sit, and in how well their encoders separate scripts. Second, head-ablation studies should check output quality. In our study, a script-change metric recorded collapsed output as a change of script and produced differences between models that disappear under a fluency check.

8 Limitations

Model scale. All four models have at most 3B parameters. Larger VLMs may organize text and image processing differently.

Table 4: Four observations are supported. The claims that the top heads route language and that head positions follow decoder family are not supported.

Observation	Evidence	Strength
Script identity is decodable from the first vision-encoder layer in all four models	Per-layer probes and a transliteration check on new images. Patch-level folds may inflate accuracy (Section 6.1).	Supported
Qwen2-VL-2B and SmolVLM use largely different heads for text and image input	Overlap above chance and stable across K , from one run (Section 6.2).	Supported
InternVL3-2B uses nearly the same heads for Hindi text and Hindi images	One model and one language (Section 6.2).	Tentative
PaliGemma-3B shares or separates heads	Overlap near its chance level (Section 6.2).	Inconclusive
Text-condition heads concentrate in layers 0–1 in the Qwen-family models	Layer counts for every language (Section 6.2).	Supported
The top 20 heads are necessary for coherent output on code-switched images	Zero ablation leaves at most 1% of outputs fluent, against 71–76% for random heads, in all four models. The fluency proxy is coarse (Section 6.3).	Supported
The top heads route output language or script	Heads of both languages cause the same collapse, and no control is matched by layer (Section 6.3).	Not supported
Shared head positions across models follow decoder family	Same-family matches are close to what layer placement predicts (Section 6.4).	Not supported

Monolingual head identification. We identify heads on monolingual sentences. RQ1 therefore compares text and image processing of single-language content, and code-switched text enters only through the probes and the ablation. Scoring heads on code-switched sentences, with the loss restricted to one language’s tokens, would link the head sets directly to code-switched input.

Ablation design. The ablations use absolute top-20 sets, a random control drawn from the whole network, and a coarse fluency proxy. These choices can show that the heads are necessary for coherent output. They cannot show what role the heads play in routing language.

Probe evaluation. The probe folds split patches, not images, so patches from one image can appear in both training and test data. This likely inflates the reported accuracy. The transliteration check uses new images and is not affected. All Devanagari patches also come from the 173 Hindi-English images among the 300 probed.

Single-run overlap. The overlap values come from one run, with the permutation baseline as the only control. Their variance across runs is unknown.

Covarying architecture. Decoder family varies together with grid shape, head count, attention type, and image compression. We cannot attribute a difference between models to any one of these factors.

Two language pairs. With two language pairs, we cannot tell whether InternVL3-2B’s Hindi-specific overlap depends on script or on language family.

9 Conclusion

We compared the attention heads that four small VLMs rely on when a sentence arrives as text and as a rendered image. The answer depends on the model. Two models use largely separate heads, with the text

heads concentrated in the first decoder layers. One model uses nearly the same heads for Hindi only, and for one model our measure is inconclusive. All vision encoders encode script identity from their first layer, so these differences do not come from a missing script signal. Ablating the top heads destroys generation in every model, while ablating random heads does not. The top heads are therefore necessary for coherent output, but the ablation cannot show that they route language. A script-change metric counts this collapse as a change of script, so head-ablation studies need a check on output quality. Future work can score heads on code-switched text, contrast heads between languages, use controls matched by layer, and test larger VLMs and language pairs that separate script from language family.

References

- Ingeol Baek, Hwan Chang, Sunghyun Ryu, and Hwanhee Lee. How do large vision-language models see text in image? Unveiling the distinctive role of OCR heads. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025. arXiv:2505.15865.
- Samyadeep Basu, Martin Grayson, Cecily Morrison, Besmira Nushi, Soheil Feizi, and Daniela Massiceti. Understanding information storage and transfer in multi-modal large language models. In *Advances in Neural Information Processing Systems*, 2024.
- Lucas Beyer, Andreas Steiner, André Susano Pinto, Alexander Kolesnikov, Xiao Wang, et al. PaliGemma: A versatile 3B VLM for transfer. *arXiv preprint arXiv:2407.07726*, 2024.
- Margaret Deuchar, Peredur Davies, Jon Russell Herring, M. Carmen Parafita Couto, and Diana Carter. Building bilingual corpora. In Enlli Môn Thomas and Ineke Mennen (eds.), *Advances in the Study of Bilingualism*, pp. 93–110. Multilingual Matters, 2014. doi: 10.21832/9781783091713-008.
- Yossi Gandelsman, Alexei A. Efros, and Jacob Steinhardt. Interpreting CLIP’s image representation via text-based decomposition. In *International Conference on Learning Representations*, 2024.
- Simran Khanuja, Sandipan Dandapat, Anirudh Srinivasan, Sunayana Sitaram, and Monojit Choudhury. GLUECoS: An evaluation benchmark for code-switched NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 3575–3585, 2020.
- Geewook Kim, Teakgyu Hong, Moonbin Yim, Jeongyeon Nam, Jinyoung Park, Jinyeong Yim, Wonseok Hwang, Sangdoo Yun, Dongyoon Han, and Seunghyun Park. OCR-free document understanding transformer. In *European Conference on Computer Vision*, 2022.
- Kenton Lee, Mandar Joshi, Iulia Raluca Turc, Hexiang Hu, Fangyu Liu, Julian Martin Eisenschlos, Urvashi Khandelwal, Peter Shaw, Ming-Wei Chang, and Kristina Toutanova. Pix2Struct: Screenshot parsing as pretraining for visual language understanding. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 18893–18912, 2023.
- Xin Liu, Qiyang Song, Qihang Zhou, Haichao Du, Shaowen Xu, Wenbo Jiang, Weijuan Zhang, and Xiaoqi Jia. Focusing on language: Revealing and exploiting language attention heads in multilingual large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. arXiv:2511.07498.
- Andrés Marafioti, Orr Zohar, Miquel Farré, Merve Noyan, Elie Bakouch, Pedro Cuenca, Cyril Zakka, Loubna Ben Allal, Anton Lozhkov, Nouamane Tazi, et al. SmolVLM: Redefining small and efficient multimodal models. *arXiv preprint arXiv:2504.05299*, 2025.
- Joanna Materzyńska, Antonio Torralba, and David Bau. Disentangling visual and written concepts in CLIP. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16410–16419, 2022.
- Clement Neo, Luke Ong, Philip Torr, Mor Geva, David Krueger, and Fazl Barez. Towards interpreting visual information processing in vision-language models. In *International Conference on Learning Representations*, 2025.

- Ercong Nie, Helmut Schmid, and Hinrich Schütze. Mechanistic understanding and mitigation of language confusion in English-centric large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025. arXiv:2505.16538.
- NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.
- Vedant Palit, Rohan Pandey, Aryaman Arora, and Paul Pu Liang. Towards vision-language mechanistic interpretability: A causal tracing tool for BLIP. In *5th Workshop on Closing the Loop Between Vision and Language, ICCV*, 2023.
- Qwen Team, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Phillip Rust, Jonas F. Lotz, Emanuele Bugliarello, Elizabeth Salesky, Miryam de Lhoneux, and Desmond Elliott. Language modelling with pixels. In *International Conference on Learning Representations*, 2023.
- Tianyi Tang, Wenyang Luo, Haoyang Huang, Dongdong Zhang, Xiaolei Wang, Xin Zhao, Furu Wei, and Ji-Rong Wen. Language-specific neurons: The key to multilingual capabilities in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024. arXiv:2402.16438.
- Michael Tschannen, Basil Mustafa, and Neil Houlsby. CLIPPO: Image-and-language understanding from pixels only. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-VL: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.
- Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. Do llamas work in English? On the latent language of multilingual transformers. *arXiv preprint arXiv:2402.10588*, 2024.
- Zekai Ye, Qiming Li, Xiaocheng Feng, Libo Qin, Yichong Huang, Baohang Li, Kui Jiang, Yang Xiang, Zhirui Zhang, Yunfei Lu, Duyu Tang, Dandan Tu, and Bing Qin. CLAIM: Mitigating multilingual object hallucination in large vision-language models with cross-lingual attention intervention. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025. arXiv:2506.11073.
- Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, et al. InternVL3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*, 2025.

A Full Overlap Values

Table 5: Jaccard overlap of the top 20 heads between conditions (A text, B image+query, C image-only). Lang is the language of the monolingual FLORES-200 sentences used to score heads. Chance levels (p_{95}) are 0.081 for Qwen2-VL-2B and InternVL3-2B, 0.053 for SmolVLM, and 0.143 for PaliGemma-3B, whose small grid raises its chance level.

Model	Lang	$A \cap B$	$A \cap C$	$B \cap C$
Qwen2-VL-2B	en	0.143	0.081	0.538
	es	0.081	0.053	0.538
	hi	0.111	0.176	0.429
InternVL3-2B	en	0.333	0.176	0.176
	es	0.538	0.250	0.212
	hi	0.905	0.667	0.667
SmolVLM	en	0.176	0.143	0.667
	es	0.111	0.111	0.905
	hi	0.250	0.212	0.905
PaliGemma-3B	en	0.250	0.333	0.333
	es	0.212	0.290	0.250
	hi	0.250	0.290	0.333

B Overlap at Other Values of K

The patterns in Section 6.2 hold at the two extremes of $K \in \{5, 10, 20, 40, 80\}$ (Table 6). The full sweep is released with the code as `checkpoints/head_overlap_matrix.json`.

Table 6: Qwen2-VL-2B and SmolVLM keep text-image overlap ($A \cap C$) below image-image overlap ($B \cap C$) at $K=5$ and $K=80$ for every language. InternVL3-2B keeps the two close for Hindi. PaliGemma-3B’s overlaps are uniformly higher because its small grid raises chance-level overlap.

Model	Lang	$A \cap C@5$	$A \cap C@80$	$B \cap C@5$	$B \cap C@80$
Qwen2-VL-2B	en	0.000	0.240	0.429	0.539
	es	0.000	0.176	0.429	0.553
	hi	0.111	0.240	0.667	0.650
InternVL3-2B	en	0.000	0.301	0.111	0.468
	es	0.000	0.301	0.000	0.416
	hi	1.000	0.584	1.000	0.739
SmolVLM	en	0.000	0.231	1.000	0.861
	es	0.111	0.203	1.000	0.839
	hi	0.000	0.368	1.000	0.882
PaliGemma-3B	en	0.250	0.495	0.429	0.778
	es	0.250	0.538	0.111	0.616
	hi	0.250	0.468	0.250	0.667

C Top Heads in Layers 0–1

Table 7: In both Qwen-family models, most text-condition heads lie in layers 0–1. Entries count how many of each top-20 set lie in those layers. Layers 0–1 hold 24 heads in Qwen2-VL-2B and InternVL3-2B, 64 in SmolVLM, and 16 in PaliGemma-3B.

Model	Lang	A: text	B: image+query	C: image-only
Qwen2-VL-2B	en	19	4	2
	es	20	3	2
	hi	20	4	6
InternVL3-2B	en	14	9	4
	es	18	15	6
	hi	18	17	14
SmolVLM	en	10	4	3
	es	14	4	4
	hi	9	5	4
PaliGemma-3B	en	5	3	6
	es	6	3	7
	hi	7	3	5

D Ablation Outcomes

Table 8 gives the fluency counts behind Figure 6. The collapsed outputs take these forms:

- PaliGemma-3B returns an empty string for 389 of 400 Hindi-head ablations.
- Qwen2-VL-2B returns output without letters for 278 of 400 Hindi-head ablations and CJK fragments for 328 of 400 English-head ablations.
- InternVL3-2B’s Hindi-head ablations contain CJK characters in all 400 generations.
- SmolVLM repeats the token `<row_2>`.

Table 8: Zero ablation of either language’s top heads leaves almost no fluent output, while random-head ablation leaves most outputs fluent. Entries count fluent outputs out of 400 (one run and three seeds of 100 images).

Model	hi-zero	en-zero	hi-mean	en-mean	random
Qwen2-VL-2B	0	0	1	1	292
InternVL3-2B	0	0	0	0	303
SmolVLM	1	1	243	3	283
PaliGemma-3B	0	4	5	0	291

Tables 9 and 10 report the script-change rates for Hindi-English, from the single run and as means over the three seeds. Nearly all zero-ablation changes are collapsed output (Section 6.3), so these rates do not measure language routing. The random-head draw in the single run changed 31% of PaliGemma-3B’s outputs, against 5–11% in the three seeded draws. About half of its generations were degenerate, which suggests that this draw hit heads the model depends on. Table 11 reports the Spanish-English rates, which count a change in the dominant language that `langid` assigns to the output.

Table 9: Script-change rates in the single run are high for Qwen2-VL-2B and PaliGemma-3B because their collapsed outputs lose all letters. Entries give the percentage of 100 Hindi-English generations whose script label changes. Columns are Qwen2-VL-2B, InternVL3-2B, SmolVLM, and PaliGemma-3B.

Variant	Qwen	Intern	Smol	PaliG
Zero-ablate en	53	25	37	18
Zero-ablate hi	74	14	37	97
Mean-ablate en	33	20	52	61
Mean-ablate hi	29	24	18	30
Random 20 heads	9	9	9	31

Table 10: The seeded runs reproduce the single-run pattern, and PaliGemma-3B’s random-head rate falls to the level of the other models. Entries give script-change rates (%) as mean $\pm$ standard deviation over three seeds.

Model	hi-zero	en-zero	random	hi-mean	en-mean
Qwen2-VL-2B	70.3 ± 5.0	46.3 ± 3.4	7.0 ± 4.9	30.3 ± 5.4	33.7 ± 8.8
InternVL3-2B	11.7 ± 1.2	17.0 ± 2.2	14.0 ± 6.2	22.7 ± 3.3	23.3 ± 2.5
SmolVLM	35.3 ± 2.1	35.3 ± 2.1	8.7 ± 0.9	23.3 ± 3.3	47.0 ± 1.6
PaliGemma-3B	97.3 ± 0.5	15.3 ± 2.9	7.0 ± 2.8	29.3 ± 2.6	65.0 ± 2.9

Table 11: Spanish-English language-change rates exceed the random-head control in every model, but they have not been checked for collapsed output. Entries give the percentage of 100 generations whose dominant language changes under zero ablation.

Model	es heads	en heads	Random 20 heads
Qwen2-VL-2B	17	14	8
InternVL3-2B	48	53	33
SmolVLM	39	51	8
PaliGemma-3B	85	82	19

E Head-Importance Maps

Figures 7 and 8 show the LAHIS scores behind the overlap analysis for Hindi and Spanish. In Qwen2-VL-2B, text-condition importance lies in layers 0–1, while the image conditions draw on heads throughout the network. In InternVL3-2B, the three conditions highlight nearly the same early heads for Hindi. SmolVLM’s importance is sparse across its wide grid, and PaliGemma-3B’s is spread across layers.

LAHIS per-head importance (Hindi), top-20 heads ringed

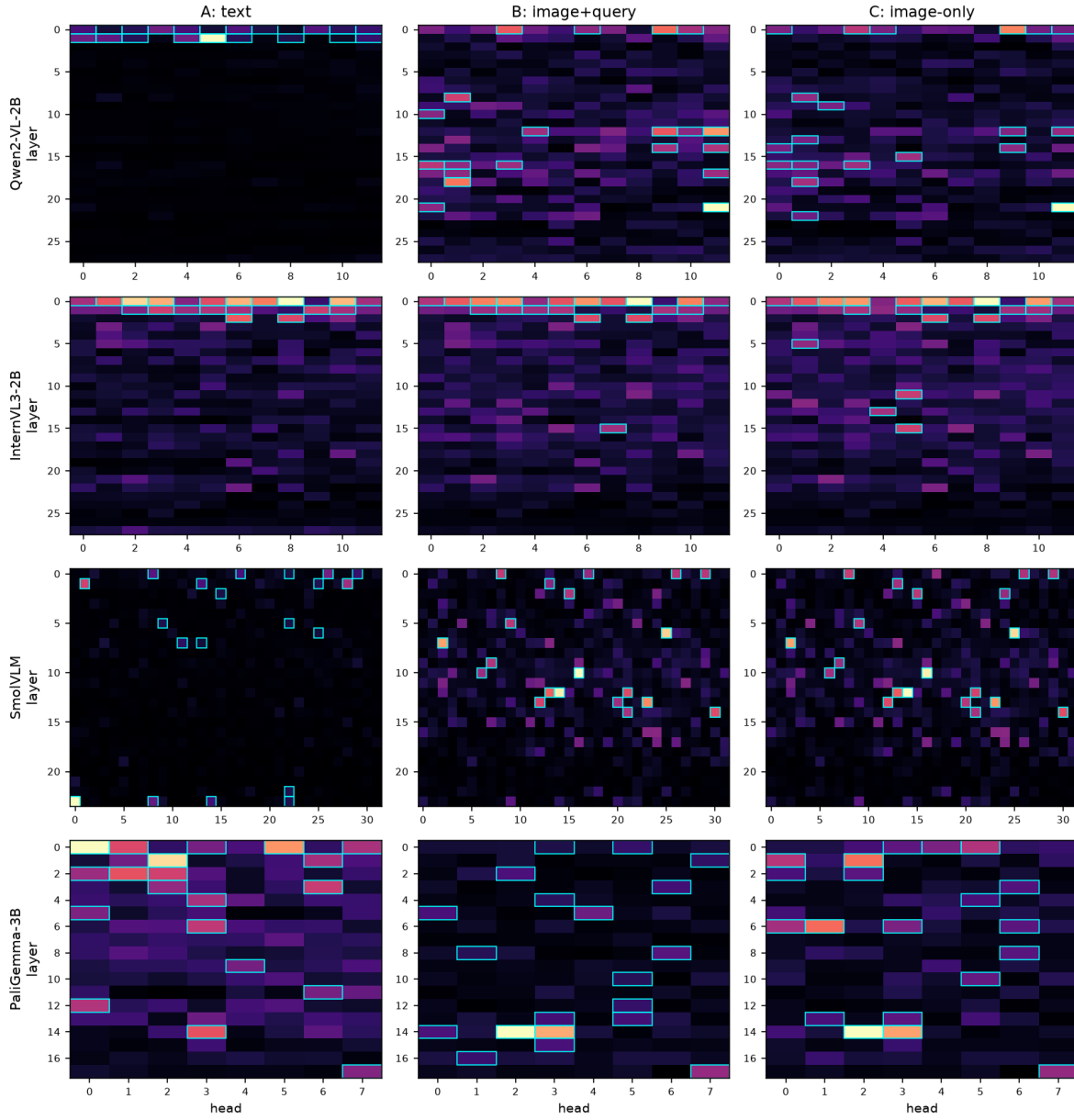


Figure 7: For Hindi, Qwen2-VL-2B’s text-condition heads lie in the first two layers while its image-condition heads spread through the network, and InternVL3-2B’s three conditions highlight the same early heads. Rows are models and columns are conditions. Cyan rings mark each panel’s top 20 heads. Scores are min-max normalized within each panel.

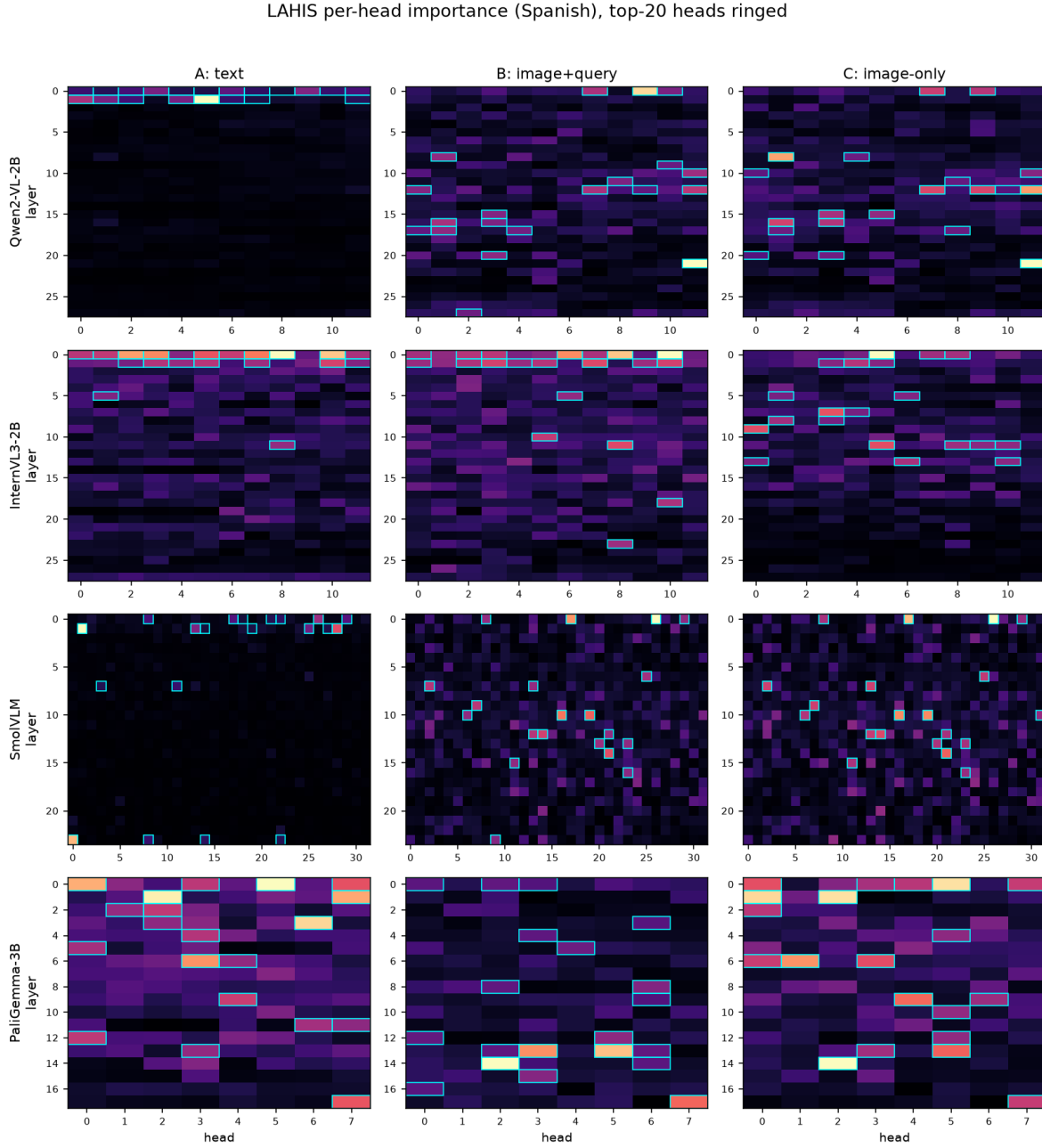


Figure 8: For Spanish, the important heads lie in the same layers as for Hindi, but InternVL3-2B’s text and image panels agree less than for Hindi. Panels follow Figure 7.